

D388

Fundamentals of Spreadsheets and Data Presentations

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

Cell B1 holds a 7% sales-tax rate. In C4 you enter $=A4*B1$ and copy it down through C20. The header row is formatted in bold blue. The results below C4 come out as zeros. Why, and what fixes it?

- A. B1 became B2, B3, and so on; use $=A4*\$B\1 so the rate cell stays fixed.
- B. Nothing is wrong; Excel always keeps references fixed when copying.
- C. Use $=\$A\$4*B1$.
- D. Change the header formatting so the formulas recalculate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Cell D2 contains $=B2*\$C\1 . What does the formula become when copied to D5?

- A. $=B5*\$C\1
- B. $=C5*\$C\1
- C. $=B5*\$C\4
- D. $=B2*\$C\1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A shop owner types $=100-20/4$ to split \$100 of profit, minus a \$20 fee, among four partners. The workbook is saved in the cloud. What does Excel return, and which formula gives what she wants?

- A. It returns 95; she needs $=100-(20/4)$.
- B. It returns 20; the formula is fine.
- C. It returns 95; she needs $=(100-20)/4$, which gives 20.
- D. It returns an error because cloud workbooks cannot divide.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

What does $=2+3^2*2$ return?

- A. 83
- B. 14
- C. 20
- D. 50

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Range A2:A11 tracks donations for a food bank's spring drive. It holds 7 numbers, 2 cells containing the text N/A, and 1 blank cell. What does $=COUNT(A2:A11)$ return?

- A. 9
- B. 10
- C. 1
- D. 7

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An inventory sheet has unsorted product IDs in column A and prices in column C. A clerk enters $=VLOOKUP(E2,A2:C500,3)$ and gets the price of a different product. The workbook has 12 tabs. What is the cause?

- A. The workbook has too many tabs.
- B. The column index should be 2.
- C. Without FALSE as the last argument, VLOOKUP uses approximate match, which needs sorted data.
- D. VLOOKUP can never work on unsorted data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher's class meets in Room 214. Students pass with a score of at least 70, and each score is in column B. Which formula in C2 labels the student correctly?

- A. $=IF(B2=70,"Pass","Fail")$
- B. $=IF(B2>70,"Pass","Fail")$
- C. $=IF(B2>=70,"Fail","Pass")$
- D. $=IF(B2>=70,"Pass","Fail")$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A sales log has region names in column A and sale amounts in column B. It has 3,000 rows and a frozen header row. Which formula totals sales for the West region?

- A. =SUM(B:B)
- B. =COUNTIF(A:A,"West")
- C. =SUMIF(B:B,"West",A:A)
- D. =SUMIF(A:A,"West",B:B)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An analyst wants to show monthly website visits over the past 24 months so leaders can spot a trend. The company's brand guide requires navy and gold. Which chart fits best?

- A. 3-D pie chart, since it is always the most engaging
- B. Line chart
- C. Pie chart
- D. Scatter plot

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which of these data sets is best shown with a pie chart?

- A. Share of a total budget across four departments
- B. Monthly revenue over five years
- C. Sales for 25 product categories
- D. Ad spend compared with sales for 50 stores

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Quarterly sales rose from \$80,000 in Q1 to \$100,000 in Q2. The sales team added two reps in Q2. What is the percent change?

- A. 125%
- B. 25%
- C. \$20,000
- D. 20%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

After a 20% increase, monthly sales are \$60,000. What were sales before the increase?

- A.** \$72,000
- B.** \$50,000
- C.** \$12,000
- D.** \$48,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

Cell B1 holds a 7% sales-tax rate. In C4 you enter $=A4*B1$ and copy it down through C20. ~~The header row is formatted in bold blue.~~ The results below C4 come out as zeros. Why, and what fixes it?

Target word: cause and fix **Decides:** The reference to B1 shifted as the formula was copied.

A	Correct	Relative references shift when copied; \$ locks the rate cell.
B	Absolute language	Relative references shift by design.
C	Inverted operation	Locks the cell that should move and leaves the rate cell free.
D	Camouflage	Formatting does not affect formula references.

2. Answer A Concept 1 · transfer

Cell D2 contains $=B2*\$C\1 . What does the formula become when copied to D5?

Target word: reverse: predict the copied formula **Decides:** Relative part shifts 3 rows; locked part stays.

A	Correct	B2 shifts down 3 rows; $\$C\1 stays locked.
B	Confusion set	Copying down changes rows, not columns.
C	Inverted operation	Shifts the locked cell and would need no \$ signs.
D	Adjacent category	Treats every reference as absolute.

3. Answer C Concept 2

A shop owner types $=100-20/4$ to split \$100 of profit, minus a \$20 fee, among four partners. ~~The workbook is saved in the cloud.~~ What does Excel return, and which formula gives what she wants?

Target word: result and fix **Decides:** Division happens before subtraction.

A	Dropped element	That is the same calculation and still returns 95.
B	Inverted operation	Reads left to right, ignoring precedence.
C	Correct	$20/4 = 5$ is done first, then $100 - 5 = 95$; parentheses force the subtraction first.
D	Camouflage	Storage location does not affect math.

4. Answer C Concept 2 · transfer

What does $=2+3^2*2$ return?

Target word: evaluate the formula **Decides:** Exponent, then multiply, then add.

A	Adjacent category	Multiplies the exponent first: $2 + 3^4$.
B	Confusion set	Treats ^ as multiplication: $2 + 3 \times 2 \times 2$.
C	Correct	$3^2 = 9$; $9 \times 2 = 18$; $2 + 18 = 20$.
D	Inverted operation	Adds first: $(2 + 3)^2 \times 2$.

5. Answer D Concept 3

Range A2:A11 tracks donations ~~for a food bank's spring drive~~. It holds 7 numbers, 2 cells containing the text N/A, and 1 blank cell. What does =COUNT(A2:A11) return?

Target word: COUNT result **Decides:** COUNT counts numbers only.

A	Confusion set	That is COUNTA: numbers plus text.
B	Dropped element	The size of the range, ignoring content type.
C	Adjacent category	That is COUNTBLANK.
D	Correct	Only the 7 numeric cells count.

6. Answer C Concept 4

An inventory sheet has unsorted product IDs in column A and prices in column C. A clerk enters =VLOOKUP(E2,A2:C500,3) and gets the price of a different product. ~~The workbook has 12 tabs~~. What is the cause?

Target word: why the wrong row returns **Decides:** The last argument was left out.

A	Camouflage	Tab count does not affect the lookup.
B	Wrong stem number	Prices are in the third column of A:C, so 3 is right.
C	Correct	Adding FALSE (or 0) forces an exact match.
D	Absolute language	Exact match works on unsorted data.

7. Answer D Concept 5

~~A teacher's class meets in Room 214~~. Students pass with a score of at least 70, and each score is in column B. Which formula in C2 labels the student correctly?

Target word: correct IF **Decides:** "At least 70" means ≥ 70 .

A	Confusion set	Passes only a score of exactly 70.
B	Dropped element	Drops the "equal to"; a score of 70 would fail.
C	Inverted operation	Swaps the true and false results.
D	Correct	Includes 70 and puts the results in the right order.

8. Answer D Concept 6

A sales log has region names in column A and sale amounts in column B. ~~It has 3,000 rows and a frozen header row~~. Which formula totals sales for the West region?

Target word: conditional total **Decides:** Sum amounts where region = West.

A	Dropped element	Drops the region condition.
B	Confusion set	Counts West rows; does not total amounts.
C	Inverted operation	Swaps the criteria and sum ranges.
D	Correct	Criteria range, criteria, then the range to add.

9. Answer B Concept 7

An analyst wants to show monthly website visits over the past 24 months so leaders can spot a trend. ~~The company's brand guide requires navy and gold.~~ Which chart fits best?

Target word: chart type **Decides:** A trend over time.

A	Absolute language	3-D distorts values and still hides the trend.
B	Correct	Lines show change across ordered time periods.
C	Confusion set	Shows parts of a whole, not trends; 24 slices would be unreadable.
D	Adjacent category	Shows relationships between two measures.

10. Answer A Concept 7 • transfer

Which of these data sets is best shown with a pie chart?

Target word: reverse: from chart to data **Decides:** Parts of one whole, few categories.

A	Correct	Few slices that add up to 100%.
B	Inverted operation	A time trend belongs in a line chart.
C	Dropped element	Drops the few-slices limit; too many to read.
D	Adjacent category	A relationship belongs in a scatter plot.

11. Answer B Concept 8

Quarterly sales rose from \$80,000 in Q1 to \$100,000 in Q2. ~~The sales team added two reps in Q2.~~ What is the percent change?

Target word: percent change **Decides:** $(\$100,000 - \$80,000) / \$80,000$.

A	Adjacent category	The ratio of new to old, not the change.
B	Correct	$\$20,000 / \$80,000 = 25\%$.
C	Dropped element	The dollar change, not a percentage.
D	Inverted operation	Divides by the new value.

12. Answer B Concept 8 • transfer

After a 20% increase, monthly sales are \$60,000. What were sales before the increase?

Target word: solve for the old value **Decides:** $\text{Old} \times 1.20 = \$60,000$.

A	Adjacent category	Applies another 20% increase.
B	Correct	$\$60,000 / 1.20 = \$50,000$.
C	Dropped element	20% of \$60,000; only a part of the value.
D	Inverted operation	Subtracts 20% of the new value.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Relative vs. absolute references	Relative references shift when a formula is copied; a \$ locks the column, row, or both (e.g., \$B\$1).	Copying a formula that points to a single rate cell without locking it.	One fixed input cell used by a formula copied down a column.
2	Order of operations	Excel evaluates parentheses, then exponents, then multiplication and division, then addition and subtraction, left to right within a level.	Reading the formula strictly left to right.	A formula mixing + or – with * or / and no parentheses.
3	COUNT vs. COUNTA vs. COUNTBLANK	COUNT counts numbers only; COUNTA counts any non-empty cell; COUNTBLANK counts empty cells.	Expecting COUNT to include text entries.	A range mixing numbers, text, and blanks.
4	Exact-match lookups	VLOOKUP needs FALSE (or 0) as its last argument for an exact match; left blank, it assumes sorted data and an approximate match.	Leaving the last argument out on unsorted ID lists.	A lookup that returns a value from the wrong row.
5	IF logic	IF(test, value_if_true, value_if_false); the comparison operator must match the rule exactly ($\geq$ vs. $>$).	Using $>$ when the rule says “at least,” or swapping the true and false results.	“At least,” “more than,” or a pass/fail rule.
6	Conditional aggregation	SUMIF(criteria_range, criteria, sum_range) adds values that meet a condition; COUNTIF counts matching cells.	Using COUNTIF for a total, or swapping the criteria and sum ranges.	“Total for only the rows where...”
7	Matching chart to message	Line for trends over time, bar or column for comparing categories, pie for parts of one whole with few slices, scatter for a relationship between two measures.	Using a pie for time trends or too many categories.	What the audience should see: trend, comparison, share, or relationship.
8	Percent change	Percent change = $(\text{new} - \text{old}) / \text{old}$.	Dividing by the new value, or reporting the ratio instead of the change.	An old value and a new value.

Course scope (public description): Fundamentals of Spreadsheets and Data Presentations offers learners an overview of the use of spreadsheet functions and methods for presenting data within spreadsheets. Learners will have the opportunity to explore features and uses of MS Excel and apply the tools to situations they may encounter while studying in their program. They will also be introduced to real world uses and tools to collect, organize and present data.